

Lightning Attention 公式推导

陆鑫

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1 Forward

将论文中的 $KV \in R^{d \times d}$ 表示为 $S \in R^{d \times d}$, 则有:

$$o_t = q_t S_t = q_t \sum_{i=1}^t k_i^\top v_i \quad (1)$$

其中 $q_t, k_t, v_t, o_t \in R^{1 \times d}$

将长度为 L 的输入序列 X 切分为长度为 C 的若干分块: $X_{[0]}, X_{[1]}, X_{[2]}, \dots$, 其中 $X_{[i]} = X_{[iC+1:(i+1)C]}$, 即令:

$$X_{[i]} = \begin{bmatrix} X_{iC+1} \\ X_{iC+2} \\ \dots \\ X_{(i+1)C} \end{bmatrix}, Q_{[i]} = \begin{bmatrix} Q_{iC+1} \\ Q_{iC+2} \\ \dots \\ Q_{(i+1)C} \end{bmatrix}, K_{[i]} = \begin{bmatrix} K_{iC+1} \\ K_{iC+2} \\ \dots \\ K_{(i+1)C} \end{bmatrix}, V_{[i]} = \begin{bmatrix} V_{iC+1} \\ V_{iC+2} \\ \dots \\ V_{(i+1)C} \end{bmatrix}$$

则:

$$X = \begin{bmatrix} X_{[0]} \\ X_{[1]} \\ \dots \\ X_{[\frac{n}{C}-1]} \end{bmatrix}, Q = \begin{bmatrix} Q_{[0]} \\ Q_{[1]} \\ \dots \\ Q_{[\frac{n}{C}-1]} \end{bmatrix}, K = \begin{bmatrix} K_{[0]} \\ K_{[1]} \\ \dots \\ K_{[\frac{n}{C}-1]} \end{bmatrix}, V = \begin{bmatrix} V_{[0]} \\ V_{[1]} \\ \dots \\ V_{[\frac{n}{C}-1]} \end{bmatrix}$$

令 $S_{[i]} \in R^{d \times d}$ 为处理完 i 个分块后的隐状态:

$$S_{[i]} = S_{iC} = \sum_{i=1}^{iC} k_i^\top v_i$$

$$S_{[i+1]} = S_{[i]} + \sum_{j=iC+1}^{(i+1)C} k_j^\top v_j = S_{[i]} + K_{[i]}^\top V_{[i]} \in R^{d \times d} \quad (2)$$

其中 $S_{[0]}$ 可以初始化为 0, 或者从之前的隐状态初始化.

则 $O_{[i+1]}$ 的计算公式为:

$$O_{[i+1]} = \underbrace{Q_{[i+1]} S_{[i+1]}}_{\text{inter-chunk: } O_{[i+1]}^{\text{inter}}} + \underbrace{((Q_{[i+1]} K_{[i+1]}^\top) \odot M) V_{[i+1]}}_{\text{intra-chunk: } O_{[i+1]}^{\text{intra}}}$$

其中 $M \in \{0, 1\}^{C \times C}$ 是掩码矩阵, 且 $M_{ij} = 1$ if $i \geq j$ else 0.

推导：

由公式 (1) 有：

$$o_t = q_t S_t = q_t \sum_{i=1}^t k_i^\top v_i$$

则：

$$\begin{aligned}
O_{[i+1]} &= \begin{bmatrix} o_{(i+1)C+1} \\ o_{(i+1)C+2} \\ \dots \\ o_{(i+2)C} \end{bmatrix} = \begin{bmatrix} q_{(i+1)C+1} \sum_{j=1}^{(i+1)C+1} k_j^\top v_j \\ q_{(i+1)C+2} \sum_{j=1}^{(i+1)C+2} k_j^\top v_j \\ \dots \\ q_{(i+2)C} \sum_{j=1}^{(i+2)C} k_j^\top v_j \end{bmatrix} \\
&= \underbrace{\begin{bmatrix} q_{(i+1)C+1} \sum_{j=1}^{(i+1)C} k_j^\top v_j \\ q_{(i+1)C+2} \sum_{j=1}^{(i+1)C} k_j^\top v_j \\ \dots \\ q_{(i+2)C} \sum_{j=1}^{(i+1)C} k_j^\top v_j \end{bmatrix}}_{\text{inter-chunk: } O_{[i+1]}^{\text{inter}}} + \underbrace{\begin{bmatrix} q_{(i+1)C+1} \sum_{j=(i+1)C+1}^{(i+1)C+1} k_j^\top v_j \\ q_{(i+1)C+2} \sum_{j=(i+1)C+1}^{(i+1)C+2} k_j^\top v_j \\ \dots \\ q_{(i+2)C} \sum_{j=(i+1)C+1}^{(i+2)C} k_j^\top v_j \end{bmatrix}}_{\text{intra-chunk: } O_{[i+1]}^{\text{intra}}} \\
O_{[i+1]}^{\text{inter}} &= \begin{bmatrix} q_{(i+1)C+1} \sum_{j=1}^{(i+1)C} k_j^\top v_j \\ q_{(i+1)C+2} \sum_{j=1}^{(i+1)C} k_j^\top v_j \\ \dots \\ q_{(i+2)C} \sum_{j=1}^{(i+1)C} k_j^\top v_j \end{bmatrix} = \begin{bmatrix} q_{(i+1)C+1} \\ q_{(i+1)C+2} \\ \dots \\ q_{(i+2)C} \end{bmatrix}_{C \times d} @ \left(\sum_{j=1}^{(i+1)C} k_j^\top v_j \right)_{d \times d} = Q_{[i+1]} S_{[i+1]} \\
O_{[i+1]}^{\text{intra}} &= \begin{bmatrix} q_{(i+1)C+1} \sum_{j=(i+1)C+1}^{(i+1)C+1} k_j^\top v_j \\ q_{(i+1)C+2} \sum_{j=(i+1)C+1}^{(i+1)C+2} k_j^\top v_j \\ \dots \\ q_{(i+2)C} \sum_{j=(i+1)C+1}^{(i+2)C} k_j^\top v_j \end{bmatrix} \\
&= \begin{bmatrix} q_{(i+1)C+1} k_{(i+1)C+1}^\top & 0 & \dots & 0 \\ q_{(i+1)C+2} k_{(i+1)C+1}^\top & q_{(i+1)C+2} k_{(i+1)C+2}^\top & \dots & 0 \\ \dots & \dots & \dots & \dots \\ q_{(i+2)C} k_{(i+1)C+1}^\top & q_{(i+2)C} k_{(i+1)C+2}^\top & \dots & q_{(i+2)C} k_{(i+2)C}^\top \end{bmatrix} @ \begin{bmatrix} v_{(i+1)C+1} \\ v_{(i+1)C+2} \\ \dots \\ v_{(i+2)C} \end{bmatrix} \\
&= \left(\begin{bmatrix} q_{(i+1)C+1} k_{(i+1)C+1}^\top & q_{(i+1)C+1} k_{(i+1)C+2}^\top & \dots & q_{(i+1)C+1} k_{(i+2)C}^\top \\ q_{(i+1)C+2} k_{(i+1)C+1}^\top & q_{(i+1)C+2} k_{(i+1)C+2}^\top & \dots & q_{(i+1)C+2} k_{(i+2)C}^\top \\ \dots & \dots & \dots & \dots \\ q_{(i+2)C} k_{(i+1)C+1}^\top & q_{(i+2)C} k_{(i+1)C+2}^\top & \dots & q_{(i+2)C} k_{(i+2)C}^\top \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} v_{(i+1)C+1} \\ v_{(i+1)C+2} \\ \dots \\ v_{(i+2)C} \end{bmatrix} \\
&= \left(\left(\begin{bmatrix} q_{(i+1)C+1} \\ q_{(i+1)C+2} \\ \dots \\ q_{(i+2)C} \end{bmatrix} @ \begin{bmatrix} k_{(i+1)C+1}^\top & k_{(i+1)C+2}^\top & \dots & k_{(i+2)C}^\top \end{bmatrix} \right) \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} v_{(i+1)C+1} \\ v_{(i+1)C+2} \\ \dots \\ v_{(i+2)C} \end{bmatrix} \\
&= ((Q_{[i+1]} K_{[i+1]}^\top) \odot M) V_{[i+1]}
\end{aligned}$$

2 Backward

根据：

$$S_{[i+1]} = S_{[i]} + K_{[i]}^\top V_{[i]} \in R^{d \times d}$$

$$O_{[i+1]} = Q_{[i+1]} S_{[i+1]} + ((Q_{[i+1]} K_{[i+1]}^\top) \odot M) V_{[i+1]} \in R^{C \times d}$$

则有：

$$dS_{[i+1]} = dS_{[i+2]} + Q_{[i+1]}^\top dO_{[i+1]}$$

$$dQ_{[i+1]} = dO_{[i+1]} S_{[i+1]}^\top + ((dO_{[i+1]} V_{[i+1]}^\top) \odot M) K_{[i+1]}$$

$$dK_{[i+1]} = V_{[i+1]} dS_{[i+2]}^\top + ((dO_{[i+1]} V_{[i+1]}^\top) \odot M)^\top Q_{[i+1]}$$

$$dV_{[i+1]} = K_{[i+1]} dS_{[i+2]} + ((Q_{[i+1]} K_{[i+1]}^\top) \odot M)^\top dO_{[i+1]}$$

推导：

由：

$$S_{[i+2]} = S_{[i+1]} + K_{[i+1]}^\top V_{[i+1]}$$

可得：

$$dS_{[i+1]}^{[1]} = dS_{[i+2]}$$

$$dK_{[i+1]}^{[1]} = V_{[i+1]} dS_{[i+2]}^\top$$

$$dV_{[i+1]}^{[1]} = K_{[i+1]} dS_{[i+2]}$$

由：

$$O_{[i+1]} = \underline{Q_{[i+1]} S_{[i+1]}} + ((Q_{[i+1]} K_{[i+1]}^\top) \odot M) V_{[i+1]}$$

可得：

$$dQ_{[i+1]}^{[1]} = dO_{[i+1]} S_{[i+1]}^\top$$

$$dS_{[i+1]}^{[2]} = Q_{[i+1]}^\top dO_{[i+1]}$$

由：

$$O_{[i+1]} = Q_{[i+1]} S_{[i+1]} + \underline{((Q_{[i+1]} K_{[i+1]}^\top) \odot M) V_{[i+1]}}$$

可得：

$$dV_{[i+1]}^{[2]} = ((Q_{[i+1]} K_{[i+1]}^\top) \odot M)^\top dO_{[i+1]}$$

$$dQ_{[i+1]}^{[2]} = ((dO_{[i+1]} V_{[i+1]}^\top) \odot M) K_{[i+1]}$$

$$dK_{[i+1]}^{[2]} = ((dO_{[i+1]} V_{[i+1]}^\top) \odot M)^\top Q_{[i+1]}$$

故：

$$dS_{[i+1]} = dS_{[i+1]}^{[1]} + dS_{[i+1]}^{[2]} = dS_{[i+2]} + Q_{[i+1]}^\top dO_{[i+1]}$$

$$dQ_{[i+1]} = dQ_{[i+1]}^{[1]} + dQ_{[i+1]}^{[2]} = dO_{[i+1]} S_{[i+1]}^\top + ((dO_{[i+1]} V_{[i+1]}^\top) \odot M) K_{[i+1]}$$

$$dK_{[i+1]} = dK_{[i+1]}^{[1]} + dK_{[i+1]}^{[2]} = V_{[i+1]} dS_{[i+2]}^\top + ((dO_{[i+1]} V_{[i+1]}^\top) \odot M)^\top Q_{[i+1]}$$

$$dV_{[i+1]} = dV_{[i+1]}^{[1]} + dV_{[i+1]}^{[2]} = K_{[i+1]} dS_{[i+2]} + ((Q_{[i+1]} K_{[i+1]}^\top) \odot M)^\top dO_{[i+1]}$$

3 Algorithm

Forward 计算公式:

$$\begin{aligned}
S_{[0]} &= 0 \in R^{d \times d} \\
S_{[i+1]} &= S_{[i]} + K_{[i]}^\top V_{[i]} \in R^{d \times d} \\
O_{[i+1]} &= Q_{[i+1]} S_{[i+1]} + ((Q_{[i+1]} K_{[i+1]}^\top) \odot M) V_{[i+1]} \in R^{B \times d}
\end{aligned}$$

Algorithm 1: Forward

Input: $Q, K, V \in R^{n \times d}$, block size B

Divide X into $T = \frac{n}{B}$ blocks $X_1, X_2, \dots, X_T \in R^{B \times d}$, $X \in \{Q, K, V, O\}$

Initialize $M \in R^{B \times B}$, $M_{ts} = \begin{cases} 1 & , t \geq s \\ 0 & , t < s \end{cases}$

Initialize $S = 0 \in R^{d \times d}$

for $t = 1, 2, \dots, T$ do

 Load $Q_t, K_t, V_t \in R^{B \times d}$ from HBM to on-chip SRAM

 On chip, compute $O^{[1]} = Q_t S$

 On chip, compute $O^{[2]} = ((Q_t K_t^\top) \odot M) V_t$

 On chip, compute $S = S + K_t^\top V_t$

 Write $O_t = O^{[1]} + O^{[2]}$ to HBM as the t-th block of O

end

return O

Backward 计算公式:

$$\begin{aligned}
dS_{[i+1]} &= dS_{[i+2]} + Q_{[i+1]}^\top dO_{[i+1]} \\
dQ_{[i+1]} &= dO_{[i+1]} S_{[i+1]}^\top + ((dO_{[i+1]} V_{[i+1]}^\top) \odot M) K_{[i+1]} \\
dK_{[i+1]} &= V_{[i+1]} dS_{[i+2]}^\top + ((dO_{[i+1]} V_{[i+1]}^\top) \odot M)^\top Q_{[i+1]} \\
dV_{[i+1]} &= K_{[i+1]} dS_{[i+2]} + ((Q_{[i+1]} K_{[i+1]}^\top) \odot M)^\top dO_{[i+1]}
\end{aligned}$$

Algorithm 2: Backward

Input: $Q, K, V, dO \in R^{n \times d}$, block size B

Divide X into $T = \frac{n}{B}$ blocks $X_1, X_2, \dots, X_T \in R^{B \times d}$, $X \in \{Q, K, V\}$

Divide dX into $T = \frac{n}{B}$ blocks $dX_1, dX_2, \dots, dX_T \in R^{B \times d}$, $X \in \{Q, K, V, O\}$

Initialize $M \in R^{B \times B}$, $M_{ts} = \begin{cases} 1 & , t \geq s \\ 0 & , t < s \end{cases}$

Initialize $S = 0 \in R^{d \times d}$, $dS = 0 \in R^{d \times d}$

for $t = 1, 2, \dots, T$ do

- Load $K_t, V_t, dO_t \in R^{B \times d}$ from HBM to on-chip SRAM
- On chip, compute $dQ^{[1]} = dO_t S^\top$
- On chip, compute $dQ^{[2]} = ((dO_t V_t^\top) \odot M) K_t$
- On chip, compute $S = S + K_t^\top V_t$
- Write $dQ_t = dQ^{[1]} + dQ^{[2]}$ to HBM as the t-th block of dQ

end

for $t = T, T-1, \dots, 1$ do

- Load $Q_t, K_t, V_t, dO_t \in R^{B \times d}$ from HBM to on-chip SRAM
- On chip, compute $dK^{[1]} = V_t dS^\top$
- On chip, compute $dK^{[2]} = ((dO_t V_t^\top) \odot M)^\top Q_t$
- On chip, compute $dV^{[1]} = K_t dS$
- On chip, compute $dV^{[2]} = ((Q_t K_t^\top) \odot M)^\top dO_t$
- On chip, compute $dS = dS + Q_t^\top dO_t$
- Write $dK_t = dK^{[1]} + dK^{[2]}$ to HBM as the t-th block of dK
- Write $dV_t = dV^{[1]} + dV^{[2]}$ to HBM as the t-th block of dV

end

return dQ, dK, dV
